

Quiz2

ⓘ This is a preview of the published version of the quiz

Started: Oct 8 at 1:08pm

Quiz Instructions

The quiz will be conducted **in person, in class**. Remote access is not permitted.

The quiz is **closed-note**.

The use of **online resources** (including search engines and generative AI) is **strictly prohibited**.

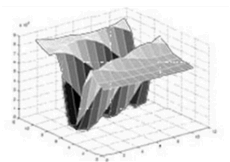
Only the Canvas quiz page should be open during the quiz. Opening any other tabs will be considered an act of academic dishonesty, and the student will be escorted out and reported.

Do not share the quiz with students who are absent, as they may need to take a make-up quiz.

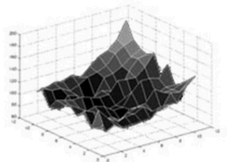
⋮
Question 1 1 pts

Correspond the three red crosses numbered 1,2,3 in the image to their autocorrelation function $E(u, v)$ shown as a surface plot:

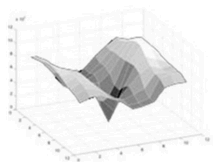
$$E(u, v) = \sum_{x, y} w(x, y) [I(x + u, y + v) - I(x, y)]^2$$



(a)



(b)



(c)

1

[Choose]

▼

2

[Choose]

▼

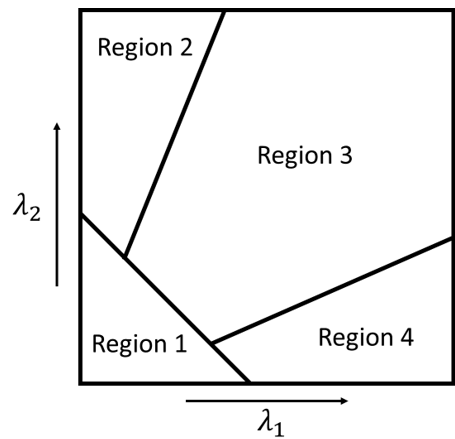
3

[Choose]

▼

⋮
Question 2 1 pts

We want to classify keypoints in image I using eigenvalues λ_1, λ_2 of $M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$. Correspond each region to its correct characteristic.



Region 1

[Choose]

F

Region 2

[Choose]

E

Region 3

[Choose]

C

Region 4

[Choose]

E

Question 3 1 pts

In the image point classification problem described above, the “cornerness” score is computed as $C = \lambda_1 \lambda_2 - \alpha (\lambda_1 + \lambda_2)^2$. Match each region shown in the previous problem to the correct cornerness value.

Region 1

[Choose]

= 0

Region 2

[Choose]

< 0

Region 3

[Choose]

> 0

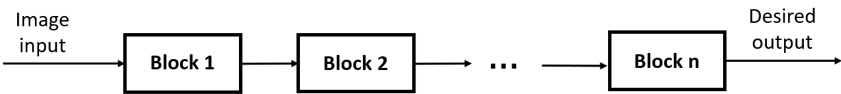
Region 4

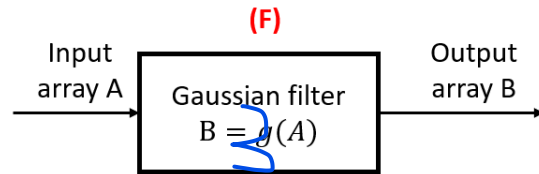
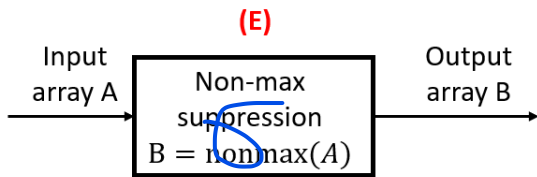
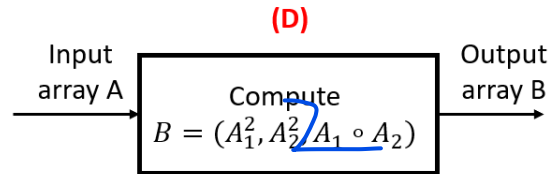
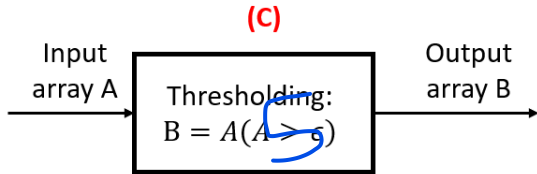
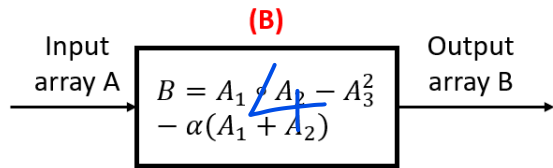
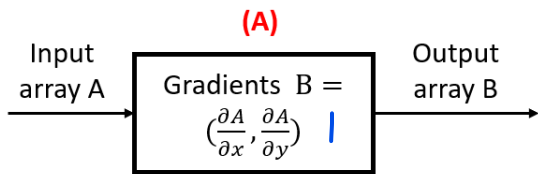
[Choose]

< 0

Question 4 1 pts

Arrange the (unordered) blocks to construct a Harris corner detector algorithm:





Block 1

Block 2

Block 3

Block 4

Block 5

Block 6

Question 5 2 pts

Given 2 images with 3 keypoints in each image, we want to match the keypoints based on the Euclidean distance between their SURF descriptors, and, using the **Lowe's threshold of 0.7** to determine whether keypoints should be matched or not. Given the table of distances between all descriptors shown below, determine the correspondences between keypoints in image 1 to image 2, and vice versa.

		Image 2		
		q_1	q_2	q_3
Image 1	p_1	10	1	100
	p_2	10	20	100
	p_3	30	100	90

p1

p2

p3

q1

q2

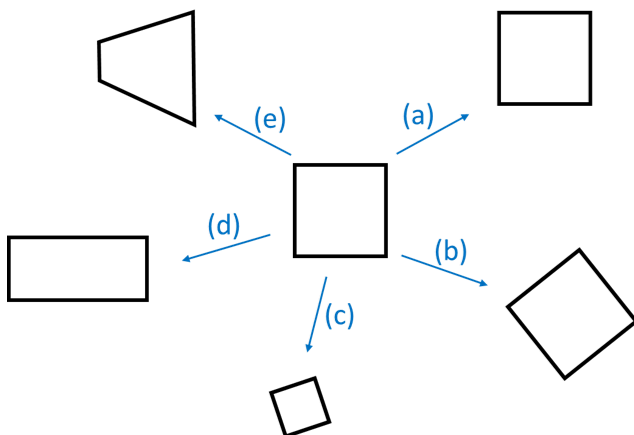
q3

[Choose]

⋮

Question 6 1 pts

Match 2D transformations based on how they transform (in the most general case) the shape of the square shown at the center:



a [Choose]

b [Choose]

c [Choose]

d [Choose]

e [Choose]

⋮

Question 7 1 pts

Recall the pinhole camera model

$$\mathbf{x} = \mathbf{P}\mathbf{X}$$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

homogeneous image coordinates camera matrix homogeneous world coordinates
3 x 1 3 x 4 4 x 1

Given the camera matrix $\mathbf{P} = \begin{bmatrix} 2 & 0 & 100 & 0 \\ 0 & 2 & 200 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, and the 3D point $\mathbf{X} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$ what is

the x-pixel coordinate of point $\mathbf{x}$ on the image? **Hint:** the x-pixel coordinate is *different* from the homogeneous image coordinate!

⋮

Question 8 1 pts

I
Rotn
Scale + Rotn
Stretch
R by 2

Match each problem to the unknown(s) that it solves.

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}$$

Unknown

Known

Known

(a)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}$$

Known

Unknown

Known

(b)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}$$

Known

Known

Unknown

(c)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}$$

Known

Unknown

Unknown

(d)

- a

[Choose]
- b

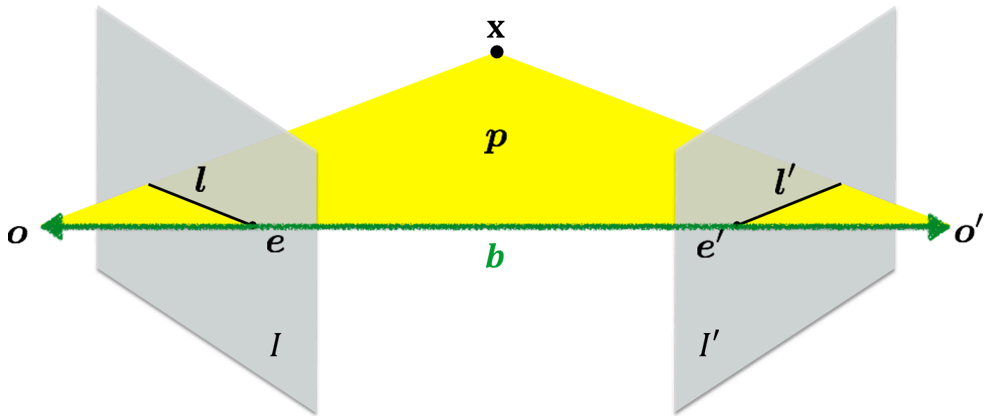
[Choose]
- c

[Choose]
- d

[Choose]

Question 9 1 pts

Correspond letters/notations to their names:



- o, o'

[Choose]
- l, l'

[Choose]
- e, e'

[Choose]
- p

[Choose]
- x

[Choose]
- b

[Choose]
- I, I'

[Choose]